Global Shifts in arXiv Preprint Production and International Co-authorship: A Country-Level Visualization Study

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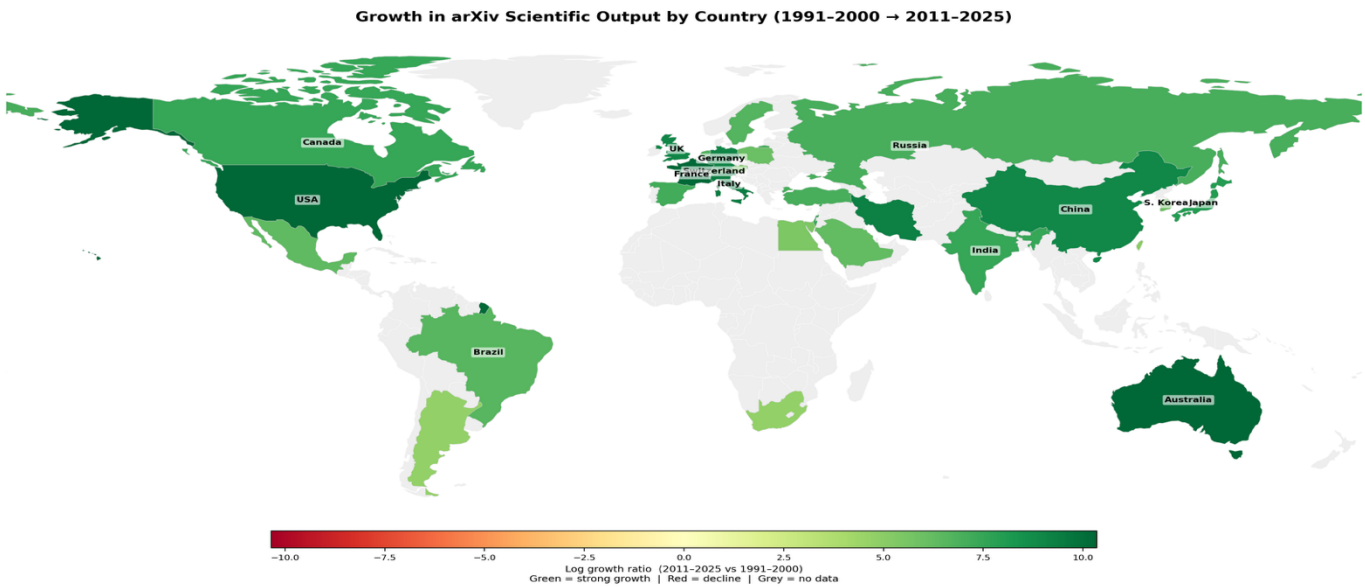
https://github.com/PomyeoLee/Data-mining-Final-project

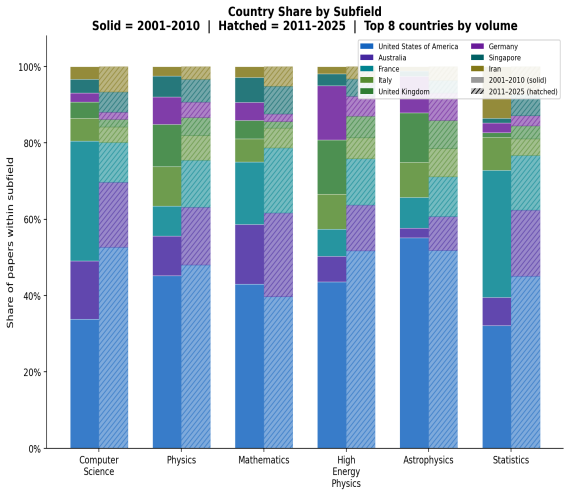
Overview

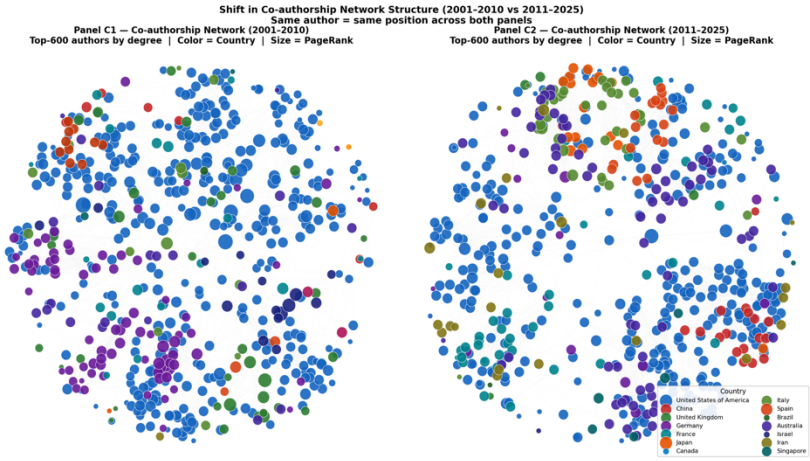
This project visualizes how scientific preprint activity on arXiv—broken down by country and research subfield—has changed across recent decades, and how co-authorship structure among highly connected authors evolved between two time windows. All figures were generated in Python (GeoPandas, Matplotlib, NetworkX) from a cleaned paper-level table derived from the public arXiv metadata snapshot.

Figures



Panel A: geographic distribution of growth in national arXiv output.





Panel B: stacked country shares within major subfields, comparing two periods.

Panel C: side-by-side co-authorship networks for two periods (same layout across panels).

Legend: visualization components

Panel A — growth choropleth (map)

Regions: Natural Earth ne_110m_admin_0_countries country polygons. **Fill color (RdYlGn colormap):** *log growth ratio* of paper counts, $\log(1 + n_{\text{recent}}) - \log(1 + n_{\text{early}})$, comparing **2011–2025** to **1991–2000**. Green indicates stronger relative growth; red indicates decline; light gray indicates no positive evidence of growth in the mapped statistic. **Color scale:** symmetric around zero (two-slope normalization using the 95th percentile of absolute log-growth for damping extremes). **Labels:** selected economies (e.g., USA, China, EU, Japan, India) annotated for orientation.

Panel B — stacked bars by subfield

Horizontal axis: six high-volume subfields (Computer Science, Physics, Mathematics, High Energy Physics, Astrophysics, Statistics). **Vertical axis:** each country's **share** of papers within that subfield and period (stacks sum to 100% within subfield/period, among plotted countries). **Stack colors:** one distinct color per country (top eight countries globally by total paper count in the pivot table). **Bar style / time encoding: solid** fills correspond to **2001–2010**; **hatched** fills correspond to **2011–2025**, shown as paired bars per subfield for direct comparison.

Panel C — co-authorship networks

Graph: undirected co-authorship: nodes are authors; edges join authors who co-authored at least one paper; edge thickness in the layout code reflects co-authorship *weight* (repeated collaboration). **Subgraph shown:** among authors with known (non-Unknown) country, the **top 600** nodes by degree in each period, for visual clarity. **Layout:** a shared **force-directed (spring)** layout computed on the *union* of the two subgraphs so the same author appears at the **same coordinates** in both panels—making structural change easier to see. **Points (nodes):** each dot is an author; **fill color** indicates inferred **country** (palette in figure legend). **Node size:** proportional to **PageRank** (weighted) within the displayed subgraph—larger nodes are more "central" in the collaboration structure for that window.

Key findings

- **Heterogeneous national growth.** Panel A highlights that relative growth in arXiv-attributed output between the early (1991–2000) and recent (2011–2025) windows is uneven across countries: some regions exhibit strong positive log-ratio growth (green), while others show stagnation or relative decline (red), revealing a shifting global footprint of open preprint dissemination.

- **Subfield–country composition shifts.** Panel B shows how the **within-subfield** concentration of activity across the top eight countries differs between 2001–2010 and 2011–2025. Solid vs. hatched stacks make period-to-period reallocations visible (e.g., whether Computer Science, Physics, or Mathematics stacks become more or less dominated by particular countries).

- **Denser collaboration and higher clustering in the recent window.** Summary statistics from the built networks indicate growth in scale (nodes/edges) from 2001–2010 to 2011–2025 alongside rising average clustering, consistent with more frequent repeat collaboration motifs in the recent cohort.

- **Network visualization underscores international mixing.** Panel C's shared layout emphasizes whether the collaboration backbone—among high-degree, country-tagged authors—rewires across periods while preserving identifiable national clusters (colors), illustrating how global preprint culture intersects with persistent author geography.

Data and methods

- **Data sources**

arXiv metadata: https://www.kaggle.com/datasets/Cornell-University/arxiv/data

Each record contributes paper identifiers, submission/update years, categories, author strings, and raw affiliation text when present.

Cartographic layer: Natural Earth `110m` administrative countries (downloaded at figure-build time if absent) for Panel A.

- **Cleaning and feature construction**

Map each paper to a **primary subfield** using the first arXiv category prefix (e.g., `cs`, `math`, `physics`) and a curated lookup table; residual labels are grouped as "Other" when needed.

Country inference for each paper combines (i) lightweight heuristics over affiliation keywords mapped to standardized country names and (ii) name/region cues; coverage is partial, so many papers remain `Unknown`—an important limitation discussed below.

Outputs include `data/papers.csv` (one row per paper with year, subfield, authors, country) and supporting maps used downstream.

Aggregations for Panels A & B

Panel A aggregates **paper counts by country** for 1991–2000 and 2011–2025 and maps the log-ratio onto country geometries after name alignment with Natural Earth.

Panel B compute each country's **share within subfield** for **2001–2010** vs. **2011–2025** for the top six subfields and eight countries.

- **Networks for Panel C**

For each period, optionally **sample up to 50,000 papers** (fixed random seed) for tractability; exclude hyper-authored papers (more than 20 authors).

Build a weighted co-authorship graph: edge weight equals the number of joint papers between author pairs. Restrict analysis to the **largest connected component (LCC)** and persist it as GraphML.

Run **Louvain** community detection (or a NetworkX greedy modularity fallback if the library is unavailable) and annotate communities; node attributes retain majority-vote `country` and subfield for exploratory analysis.

Panel C reads these GraphML files, applies the shared spring layout, and encodes PageRank and country as described above.

Significance

Open preprint archives such as arXiv have become central infrastructure for global research communication. Mapping **national participation**, **disciplinary specialization**, and **collaboration topology** makes systemic shifts visible—including where production grows fastest, how disciplinary arenas redistribute across countries, and how collaboration networks thicken over time. Such evidence matters for science policy, equitable access to visibility in open systems, and understanding the geography of mathematical and quantitative disciplines. Transparent methods (full pipeline in Python), explicit geographic and network caveats, and a permanent GitHub archive support reproducibility and classroom critique of inference from large bibliometric traces.